

Task: Build an SOP Operations Dashboard from 25 API Endpoints

Assigned by: Charan, Founder & CEO, ElevateBox **Role context:** You are being trained to work in a Founder's Office for a business consultant who drafts SOPs for companies. The dashboards you build will track SOP metrics, trigger SOP personnel when thresholds are hit, and help audit whether services were delivered on time.

What You Are Building

A single, production-grade dashboard that pulls live data from 25 different API endpoints. The dashboard should have charts, tables, KPI cards, and alerts, all updating from real data sources.

Every widget on this dashboard must answer one of three questions:

1. Should someone be triggered to execute an SOP right now?
2. Did the team execute the SOP within the agreed SLA?
3. Is the client's business environment changing enough to warrant an SOP refresh?

If a widget cannot answer one of these, it does not belong on the dashboard.

Tech Stack (Recommended)

- **Backend:** Node.js or Python (FastAPI/Flask)
 - **Frontend:** React/Next.js or Streamlit
 - **Charts:** Recharts, Plotly, or Apache ECharts
 - **Cache:** SQLite or Redis (mandatory, not optional)
 - **Scheduling:** Cron jobs or GitHub Actions for periodic data refresh
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The 25 Endpoints

CATEGORY A: Public APIs (No Authentication) — 8 Endpoints

A1. CoinGecko (Crypto Prices)

- Endpoint:
https://api.coingecko.com/api/v3/simple/price?ids=bitcoin,ethereum&vs_currencies=usd
- Auth: None
- Rate limit: ~30 calls/min with free demo key
- Build: KPI cards for top 5 currencies + 24hr sparkline
- SOP trigger: If any client treasury asset moves >10% intraday, trigger "Treasury Reserve Review SOP"
- Docs: <https://docs.coingecko.com/reference/endpoint-overview>

A2. Frankfurter (Foreign Exchange Rates)

- Endpoint:
<https://api.frankfurter.dev/v1/latest?from=USD&to=EUR,GBP,INR>
- Auth: None
- Rate limit: No documented quota
- Build: Line chart of 30-day multi-currency trend + today's rate KPI strip
- SOP trigger: If USD to client currency moves >2% week-over-week, trigger "Cross-border Invoicing SOP"
- Docs: <https://frankfurter.dev>

A3. World Bank Indicators (Country Macro Data)

- Endpoint:
<https://api.worldbank.org/v2/country/IND/indicator/NY.GDP.MKTP.CD?format=json>
- Auth: None
- Rate limit: Etiquette is 1 req/sec
- Build: Bar chart comparing client's HQ country vs operating geographies on GDP, inflation, ease of doing business
- SOP trigger: If a country's inflation deviates >2 standard deviations from its 10-year mean, flag "Pricing Review SOP"
- Docs: <https://datahelpdesk.worldbank.org/knowledgebase/articles/889392>

A4. Hacker News (Top Stories)

- Endpoint: <https://hacker-news.firebaseio.com/v0/topstories.json> then [/v0/item/{id}.json](https://hacker-news.firebaseio.com/v0/item/{id}.json) for each story
- Auth: None
- Rate limit: No official limit, but cache aggressively
- Build: Table of top 20 stories with score/comment count + bar chart of stories per domain
- SOP trigger: If client's domain appears on HN front page, trigger "Inbound Press Response SOP"
- Docs: <https://github.com/HackerNews/API>

A5. WHO Global Health Observatory (Health Indicators)

- Endpoint: <https://ghoapi.azureedge.net/api/NCDMORT3070> (premature NCD mortality as example)
- Auth: None
- Format: OData JSON
- Build: Heatmap of public health indicators by country
- SOP trigger: If a tracked health indicator worsens quarter over quarter, flag for "Compliance Audit SOP"
- Docs: <https://www.who.int/data/gho/info/gho-odata-api>

A6. Open-Meteo (Air Quality Forecast)

- Endpoint: https://air-quality-api.open-meteo.com/v1/air-quality?latitude=17.385&longitude=78.4867&hourly=pm2_5,pm10,ozone
- Auth: None
- Build: Gauge for current AQI per office location + line chart of 48hr pollution forecast
- SOP trigger: If AQI > 200 at a client's office city, trigger "WFH Advisory SOP"
- Docs: <https://open-meteo.com/en/docs/air-quality-api>

A7. RandomUser.me (Mock HR Data)

- Endpoint: <https://randomuser.me/api/?results=50&nat=us,in,gb>
- Auth: None
- Build: HR-style table with gender, age, country distribution + avatar grid
- Purpose: Seed the HR section of the dashboard during development. Later, swap this for real Airtable/Notion HRIS data without rewriting the visualizations.
- Docs: <https://randomuser.me/documentation>

A8. Reddit Public JSON (Community Sentiment)

- Endpoint:
<https://www.reddit.com/r/Entrepreneur/top.json?limit=25&t=day>
 - Auth: None (set a descriptive User-Agent header to avoid 429s)
 - Build: Word cloud of top post titles in client-relevant subreddits + table of top complaints
 - SOP trigger: If a competitor's subreddit has a complaint spike (>2x rolling average), trigger "Competitive Intelligence Briefing SOP"
 - Docs: <https://www.reddit.com/dev/api/>
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CATEGORY B: Private / Key-Based APIs (Free Registration Required) — 10 Endpoints

For each of these, you must register, get your API key, and store it in an `.env` file. Never commit keys to git.

B1. Alpha Vantage (Stock & FX Data)

- Endpoint:
https://www.alphavantage.co/query?function=GLOBAL_QUOTE&symbol=RELIANCE.BSE&apikey=YOUR_KEY
- Auth: API key in query string
- Get key: <https://www.alphavantage.co/support/#api-key>
- Rate limit: 25 requests/day on free tier (this is the binding constraint, plan accordingly)
- Build: Candlestick chart for top 5 client portfolio tickers + KPI card for daily change
- SOP trigger: If a tracked equity moves >5%, trigger "Investor Update SOP"
- Docs: <https://www.alphavantage.co/documentation/>

B2. OpenWeatherMap (Current Weather + Forecast)

- Endpoint:
https://api.openweathermap.org/data/2.5/weather?lat=17.385&lon=78.4867&appid=YOUR_KEY&units=metric
- Auth: API key in query string (`appid=`)
- Get key: https://home.openweathermap.org/users/sign_up
- Rate limit: 60 calls/min, ~1M calls/month
- Build: Multi-city KPI strip (temperature, humidity, wind, weather icon)
- SOP trigger: If severe weather alerts are present, trigger "Business Continuity SOP"
- Docs: <https://openweathermap.org/api>

B3. NewsAPI (Headlines & Articles)

- Endpoint:
https://newsapi.org/v2/top-headlines?country=in&category=business&apiKey=YOUR_KEY
- Auth: API key in header (X-Api-Key) or query string
- Get key: <https://newsapi.org/register>
- Rate limit: 100 requests/day, 24hr article delay on free plan
- Build: News ticker/table filtered by client industry + bar chart of mentions per day
- SOP trigger: If client name appears with negative sentiment keywords, trigger "Crisis Comms SOP"
- Docs: <https://newsapi.org/docs>

B4. FRED (Federal Reserve Economic Data)

- Endpoint:
https://api.stlouisfed.org/fred/series/observations?series_id=CPIAUCSL&api_key=YOUR_KEY&file_type=json
- Auth: API key in query string
- Get key: https://fred.stlouisfed.org/docs/api/api_key.html
- Build: Line chart of CPI / Unemployment / Yield Curve + KPI card with month-over-month change
- SOP trigger: If CPI shifts >0.3% MoM, trigger "Pricing Review SOP"
- Docs: <https://fred.stlouisfed.org/docs/api/fred/>

B5. USAJOBS (Federal Job Postings)

- Endpoint:
<https://data.usajobs.gov/api/Search?Keyword=compliance&LocationName=Washington,DC>
- Auth: API key in Authorization-Key header + User-Agent set to your registration email
- Get key: <https://developer.usajobs.gov/APIRequest/Index>
- Build: Table of active job openings + bar chart of openings by agency
- SOP trigger: For govcon clients, new relevant postings auto-trigger "Capture Management SOP"
- Docs: <https://developer.usajobs.gov/api-reference/>

B6. Clockify (Time Tracking)

- Endpoint: GET
<https://api.clockify.me/api/v1/workspaces/{workspaceId}/time-entries>
- Auth: X-Api-Key custom header
- Get key: Clockify Profile Settings > API section

- Rate limit: 30 requests/hour on free plan
- Build: Stacked bar chart of hours per project per consultant + KPI cards for billable utilization %
- SOP trigger: If utilization >90% for 3 consecutive weeks, trigger "Capacity Reallocation SOP"
- Docs: <https://docs.clockify.me/>

B7. Notion API (SOP Registry Database)

- Endpoint: POST
https://api.notion.com/v1/databases/{database_id}/query
- Auth: Bearer token in **Authorization** header + **Notion-Version: 2022-06-28**
- Get key: Create an internal integration at <https://www.notion.so/my-integrations>
- Rate limit: Average 3 requests/second
- Build: Kanban-style flowchart showing SOP statuses (Draft > Review > Published > Retired)
- SOP trigger: If any SOP has not been reviewed in 6+ months, flag for "SOP Refresh Workflow"
- Docs: <https://developers.notion.com/reference/intro>

B8. Airtable (Client CRM / Roster)

- Endpoint: GET <https://api.airtable.com/v0/{baseId}/{tableName}>
- Auth: Bearer token (Personal Access Token) in **Authorization** header
- Get key: <https://airtable.com/create/tokens>
- Rate limit: 5 requests/second per base
- Build: Pivot table of client roster + gauge for SOP coverage % (SOPs delivered / total engagements)
- SOP trigger: If onboarding checklist rows are incomplete past 7 days, auto-escalate
- Docs: <https://airtable.com/developers/web/api/>

B9. Trello (Task/Workflow Boards)

- Endpoint: GET
<https://api.trello.com/1/members/me/boards?key=KEY&token=TOKEN>
- Auth: API Key + Token pair in query params (this is OAuth 1.0 under the hood)
- Get key: <https://trello.com/power-ups/admin> then API Key section
- Rate limit: 300 req per 10s per API key
- Build: Sankey diagram of cards by list (where work sits in the pipeline) + burn-down chart of completed cards per week
- SOP trigger: If a card stays in "Blocked" for >3 days, trigger escalation
- Docs: <https://developer.atlassian.com/cloud/trello/rest/>

B10. AQICN (World Air Quality Index)

- Endpoint: https://api.waqi.info/feed/hyderabad/?token=YOUR_KEY
 - Auth: API key in query string (`token=`)
 - Get key: <https://aqicn.org/data-platform/token/>
 - Rate limit: 1,000 req/second (very generous)
 - Build: Heatmap of city-level AQI across client office locations + alert card when AQI > 150
 - SOP trigger: Same WFH advisory family as Open-Meteo, but AQICN gives station-measured (audit-grade) data
 - Docs: <https://aqicn.org/api/>
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CATEGORY C: Website Scraping (No Formal API) — 7 Endpoints

For every scraper: set a descriptive `User-Agent` header (e.g., `SOP-Audit-Bot/1.0 yourname@email.com`), respect `robots.txt`, and keep requests to 1 per 2-5 seconds.

C1. SEC EDGAR (US Company Filings)

- URL: <https://data.sec.gov/submissions/CIK0000320193.json> (Apple example)
- Also:
<https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK=0000320193&type=10-K&count=40>
- Hard rule: Max 10 req/sec, `User-Agent` must contain your name + email or you get blocked
- Build: Timeline of client filings + alert card for new 8-K filings
- SOP trigger: New 8-K filing triggers "Material Event Memo SOP" with a 24hr SLA
- Docs: <https://www.sec.gov/search-filings/edgar-search-assistance/accessing-edgar-data>

C2. Hacker News "Who Is Hiring" Thread (Hiring Signals)

- URL: Find latest thread via <https://news.ycombinator.com/submitted?id=whoishiring>, then scrape comments or use Firebase API for each comment
- Respect: `robots.txt` sets `Crawl-delay: 30`
- Build: Bar chart of hiring volume by month/tech stack, filterable by client portfolio companies
- SOP trigger: If a client startup is hiring, trigger "Org Design Refresh SOP"

C3. RemoteOK (Remote Job Listings)

- URL: <https://remoteok.com/api> (returns JSON, first element is a legal notice you must respect)
- Also scrapeable at: <https://remoteok.com/> (each listing is `tr.job`)
- Build: Bubble chart (X = posting date, Y = salary, size = applications)
- SOP trigger: "Market Salary Benchmark SOP" pulls last 30 days to refresh comp ranges quarterly

C4. Wikipedia Company Infoboxes (Client Profiles)

- URL: https://en.wikipedia.org/wiki/Stripe,_Inc. (example)
- Selector: `table.infobox.vcard`
- Also consider:
https://en.wikipedia.org/api/rest_v1/page/summary/{title} (REST API, cleaner)
- Required: User-Agent with contact info, max 1 req/sec per Wikimedia policy
- Build: KPI tile grid for each client (founded, HQ, revenue, employee count, leadership)
- SOP trigger: Leadership change triggers "Re-introduction Call SOP"

C5. Yahoo Finance (Stock Ticker Pages)

- URL: <https://finance.yahoo.com/quote/AAPL>
- Caveat: Partially JS-rendered. Price values are in initial HTML, but financials tab needs a headless browser (Playwright). Use for training only, not production.
- Build: Candlestick + news feed combo card
- SOP trigger: >5% intraday move triggers "Market Event Memo SOP"

C6. Wikipedia Pageviews API (Brand Attention Tracker)

- URL:
https://wikimedia.org/api/rest_v1/metrics/pageviews/per-article/en.wikipedia.org/all-access/user/{Article_Title}/daily/{YYYYMMDD}/{YYYYMMDD}
- Auth: None, but set User-Agent per Wikimedia policy
- Build: Line chart of client vs competitor weekly pageviews + anomaly flag when views > 2x 30-day rolling mean
- SOP trigger: A spike triggers "Reputation Audit SOP"
- Docs:
<https://doc.wikimedia.org/generated-data-platform/aqs/analytics-api/reference/page-views.html>

C7. India MCA Company Data (via data.gov.in bulk CSV)

- URL: Browse <https://www.data.gov.in/catalog/company-master-data> and download the latest CSV per ROC
 - Do NOT scrape MCA portal directly (it uses CAPTCHA + session tokens)
 - Build: Filterable table of client legal entities + alert card when status flips to "Under Strike Off" or "Dormant"
 - SOP trigger: Status change or director change triggers "KYC/Compliance Refresh SOP"
 - Important lesson: OpenCorporates explicitly prohibits scraping in their ToS. Read ToS before scraping anything. This is not optional.
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Submission Checkpoints

Checkpoint 1 (End of Week 1): Public API Layer

- ☐ All 8 public APIs (A1 through A8) are integrated and returning live data
- ☐ Each endpoint has a working visualization on the dashboard (correct chart type as specified above)
- ☐ A SQLite or Redis cache layer is in place with TTLs (no raw API calls on every page load)
- ☐ Every widget renders within 2 seconds using stale-while-revalidate pattern
- ☐ You can verbally explain in 60 seconds how each widget maps to a specific SOP trigger
- **Deliverable:** Live demo + code pushed to git

Checkpoint 2 (End of Week 2): Key-Based API Layer

- ☐ All 10 key-based APIs (B1 through B10) are integrated
- ☐ All API keys are stored in `.env` file, never committed to git (`.gitignore` entry exists)
- ☐ A per-provider retry/backoff wrapper exists (not copy-pasted retry logic in every file)
- ☐ You can deliberately trigger a 429 from Alpha Vantage and show the system degrades gracefully (cached value + visible "stale data" badge on the widget)
- ☐ Auth patterns are correct: query string keys for Alpha Vantage/OpenWeather/FRED, custom header for Clockify/USAJOBS, Bearer token for Notion/Airtable, Key+Token pair for Trello
- **Deliverable:** Live demo showing graceful degradation + code review with senior dev

Checkpoint 3 (End of Week 3): Scraping Layer

- ☐ All 7 scraping targets (C1 through C7) are integrated
- ☐ Every scraper has a YAML config file committed to git containing: User-Agent string, robots.txt evaluation, rate limit policy, and CSS selectors used

- ☐ A senior reviewer can pause any scraper by editing the YAML config (no code change needed)
- ☐ SEC EDGAR requests include proper User-Agent with name + email
- ☐ You have documented one example where a website's ToS prohibits scraping (OpenCorporates) and explained what you would do instead
- **Deliverable:** Code review focused on scraper architecture + YAML configs

Checkpoint 4 (End of Week 4): SOP Trigger Integration + Final Dashboard

- ☐ An "SOP Trigger Rules" table exists (in Notion or Airtable) with the structure: (metric, comparator, threshold) > (SOP_ID, assignee, SLA_hours)
- ☐ At least 15 of the 25 widgets have a working trigger rule that creates an action item when the threshold is hit
- ☐ A single "SOP Action Queue" view exists on the dashboard that the founder can scan in under 60 seconds each morning
- ☐ You demo a full loop: a real data event (e.g., a tracked ticker drops 6%) flows into the trigger table and creates a card in Trello with the correct SOP attached
- ☐ One-pager documentation exists for each of the 25 endpoints covering: auth method, rate limit, retry policy, response schema, and which SOP it powers
- ☐ The full dashboard is deployed and accessible via a URL (Vercel, Railway, or similar)
- **Deliverable:** Final demo to Charan + deployed URL + 25 one-pagers in a [/docs](#) folder

Rules

1. No API key or secret should ever appear in git history. If it does, rotate the key immediately.
 2. Every API call must go through a cache layer. No exceptions.
 3. Every scraper must respect robots.txt and rate limits. If you get a 429, your backoff logic should handle it automatically.
 4. If a free tier runs out or an API changes, swap to the documented fallback (e.g., Alpha Vantage > Twelve Data or yfinance, NewsAPI > GNews or NewsData.io). Structure your code so swapping a provider is a single-file change.
 5. Write clean, readable code. If someone else cannot understand your API integration file in 5 minutes, rewrite it.
 6. Each widget must have a visible "Last Updated" timestamp so the founder knows if data is stale.
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Fallback Providers (If a Free Tier Breaks)

Primary	Fallback	Reason
Alpha Vantage	Twelve Data or yfinance library	25/day cap is very tight
NewsAPI	GNews (100/day) or NewsData.io (200 credits/day)	24hr delay on free NewsAPI
Trello (OAuth complexity)	Use long-lived Key+Token pair for training, defer full OAuth to later	
WHO GHG OData	Cache heavily, API is in transition	Schema may change
Yahoo Finance scraping	Already have Alpha Vantage / CoinGecko for price data	Yahoo ToS restricts automated access

What You Will Learn

By the end of this task, you should be able to:

- Hit any API endpoint using Postman, curl, or code and understand the response
- Handle five different auth patterns (none, query key, custom header, Bearer token, OAuth key+token)
- Build a cache layer that prevents you from burning through free tier limits
- Write retry/backoff logic that handles 429s and 5xx errors gracefully
- Choose the right chart type for the data shape (KPI card for single numbers, line chart for time series, bar chart for comparisons, heatmap for geographic data, gauge for thresholds, table for detail views, Sankey for flow)
- Scrape a website responsibly (User-Agent, robots.txt, rate limits, ToS awareness)
- Connect a data point to a business action (the "if this metric crosses this threshold, trigger this SOP for this person" logic)
- Deploy a full-stack dashboard that a non-technical founder can use every morning

This is the exact skill set needed to build dashboards in a founder's office where data drives SOP execution and service delivery audits.

